Week 39 Road detours

Facilitator guide keyed to the revised student packets

Mathematical purpose

A journey is an ordered finite list of directed traversals of actual edges on a fixed graph. The only shortening move deletes two adjacent traversals of the same edge in opposite directions. The graph, start, and finish remain fixed. The empty journey means staying at the initial vertex.

Unique reduction. Every journey ends in the same reduced edge path, regardless of deletion order. Each deletion shortens by two, so the process terminates. A running stack of oriented edges gives an invariant normal form; the proof is on page 6. “Same endpoints” is enough to identify an edge only on these simple printed graphs. Parallel edges would need separate road identities.

Tree theorem. Between two vertices of a tree there is a unique simple path, and every journey between them reduces to it. In particular, every closed tree journey reduces to the empty journey. A reduced path on a four-edge ring, by contrast, keeps going in a single direction and may survive for one or more complete circuits.

Order can survive. Let a be H→A→B→H and b be H→C→D→H on the two-ring map. The journeys ab and ba have distinct reduced paths. The commutator ab a^{-1} b^{-1} has zero net count around each ring but has no adjacent inverse edges, so it does not disappear. Net traversal counts lose essential order information.

Topological scope. These elementary edge moves describe endpoint-fixed homotopy of graph edge paths, with immediate inverse-pair insertions allowed when comparing equivalence classes. The student shortening activity uses deletions. The open regions inside rings are not filled faces: paths cannot slide across them, cancel separated inverse steps, trade subjourney order, or move their basepoint. Paths may revisit vertices and overlap themselves.

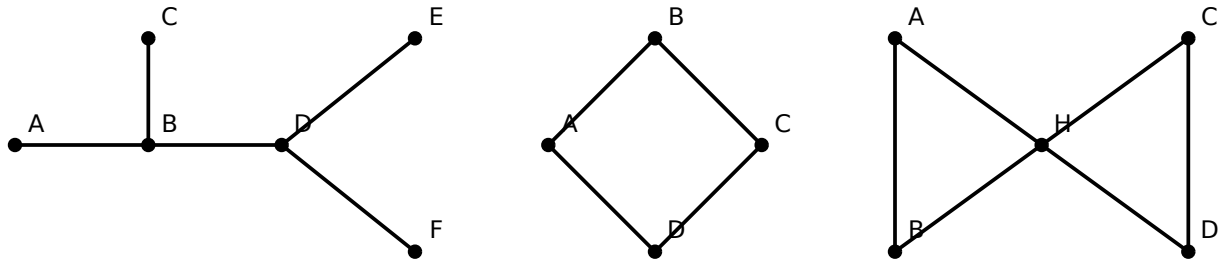
Entry points and readiness

K-1: play and replay short journeys with a pawn; an adult records ordered steps. **Grades 2-3:** construct long tree walks, classify ring results, and contrast order on two rings. **Grades 4-5:** prove unique reduction and find a zero-net-count trip that cannot disappear. Counting to 12 is useful, not fluent reading or algebra.

Before-use gate: the exact large route tiles, adult-recording method, mats, and pawn workflow have not been physically tried with the intended group. A K-1 child must be able to replay an ordered route and remove an immediate reversal without changing the remaining order. Stay with the concrete demonstration if that is not yet reliable.

Preparation demonstration and a flexible hour

Materials. Per child: a pawn; the printed maps on the student pages are the mats. Per pair: sixteen blank erasable step cards at least 30 mm long; write both endpoint labels and an arrow on one card for each actual traversal, including repeats. Each table also has twenty spare cards, sleeves and pens. Five pair-kits suit the current 4/4/3 children with three anchored adults; at the upper table rotate the third child as recorder or referee. Adults prepare cards before use. Use the six-page K-1 packet and the four-page Grades 2-3 and 4-5 packets. The K-1 key, Problems 1-7, is on page 3.



Exact map identities. Tree edges: AB, BC, BD, DE, DF. Ring: AB, BC, CD, DA. Two-ring map: HA, AB, BH and HC, CD, DH. The graph is never erased. Every recorded step identifies a particular road and direction; physical tile arrows without a road identity are insufficient when a junction offers several choices.

Before-use test. Replay the packet's A→B→C→B→D example with your pawn and actual tiles. Remove the middle B→C and C→B tiles, then replay A→B→D. Check that the child sees deletion of a journey excursion rather than removal of a road. Confirm the route remains readable when tiles are lifted and that the empty result can be recorded as "stay at A." This practical trial remains unperformed here.

Launch, 5-7 minutes. First let everyone travel freely. Show four step tiles in order for A→B→C→B→D. Point to the two adjacent opposite traversals of BC, remove just those tiles, and replay the surviving route. Use the same labels in input, process, and output, as on student page 1. Do not teach the stack algorithm before children explore removal choices.

Time	Suggested work
0-7 min	Free pawn exploration and the four-step removal demonstration.
7-23 min	K-1 Problems 1-2: short tree journeys and four-step ring contrast; adults record.
23-38 min	Continue K-1 short journeys; older children can compare deletion orders and ring paths.
38-51 min	K-1 Problems 3-7 only when ready, or the older two-ring challenges; save unfinished work.
51-60 min	Compare a conjecture with a reason; share one route that survives.

Pacing and support. Use fewer tasks rather than faster transcription. Ask a child to replay what remains before accepting a cancellation. If order recording overwhelms the youngest child, use one short adult-recorded route; do not turn this into a worksheet of letter-string manipulation. No cutting or rope tightening is needed; any materials trimming is adult-only.

K-1 answer key

In this key a compact route such as ABDE means A→B→D→E. Count edges, not letters: a seven-step route has eight displayed vertices. The starting letter remains when every step is removed.

Problem 1. The three-step A-to-E trip is A-B-D-E. Every four-step B-home trip reduces to staying at B; every four-step A-to-D trip reduces to A-B-D. For example B-A-B-A-B disappears, while A-B-A-B-D leaves A-B-D.

Problem 2. The possible reduced four-step ring home results are stay at A, A-B-C-D-A, and A-D-C-B-A. A nonempty reduced ring trip keeps one direction; the two circuits survive.

Problem 3. Example seven-step A-to-E trip: ABABDFDE, reducing to ABDE. Example seven-step C-to-F trip: CBABDEDF, reducing to CBDF. Any valid constructions have these same respective reduced routes, the unique simple paths in the tree.

Problem 4. All 4-, 6-, 8-, and 12-step trips from A back to A shorten to staying at A. Explicit witnesses alternate A and B for 2, 3, 4, or 6 out-and-back pairs. Encourage less repetitive excursions when children are ready; the theorem covers all valid choices.

Problem 5. CBABDEDBDF has nine steps, uses AB, BC, BD, DE, and DF, and ends at F. Delete B→A→B, D→E→D, and the later D→B→D excursion as they become adjacent to get CBDF. No two can end differently: every trip from C to F shortens to C→B→D→F.

Problem 6. There are exactly five reduced home results from eight-step ring walks:

Remaining route	Length	One eight-step input
Stay at A	0	ABABABABA
ABCDA	4	ABABABCDA
ADCBA	4	ABABADCBA
ABCDABCDA	8	ABCDABCDA
ADCBADCBA	8	ADCBADCBA

Completeness: on a degree-two ring, once a nonempty reduced route chooses a direction it cannot turn around. A home route has length a multiple of four. Deletions preserve parity and cannot increase length; among lengths 0 through 8 this gives 0, 4, or 8, with two directions for each nonzero length. The listed inputs prove attainability.

Problem 7. Exactly four results: ABC, ADC, ABCDABC, and ADCBADC. Their lengths are 2, 2, 6, and 6. Six-step inputs for the short results include ABABABC and ABABADC. A nonreversing route from A to the opposite vertex C has length 2 modulo 4, so no other result of length at most 6 exists. Yes, the two six-step survivors both use all four roads, but in different orders. They do not have the same directed-edge multiset; the question only asks for the same roads. Accept no with the reason that the tiles differ.

Grades 2-3 answer key

Problem 1. The three printed routes reduce, in order, to ABDE, ABDE, and CBDF. One different seven-step route from C to F is CBABDEDF. Its two removable excursions are B→A→B and D→E→D; what remains is CBDF.

Problem 2. Valid 12-step A-home trips covering every road include ABABCBDEDFDBA, ABABCBDFDEDBA, and ABABDEDFDBCBA. Every one reduces to A. A nonempty reduced closed walk in a tree would force a cycle; equivalently a farthest excursion forces a reversal. The claim holds for all finite lengths, not just these examples.

Problem 3. The complete eight-step ring result set is A (stay), ABCDA, ADCBA, ABCDABCDA, and ADCBADCBA. The fewest steps in a nonempty surviving trip home is four. A degree-two ring leaves a reduced walker no choice except continuing in its chosen direction; a closed route must complete full four-edge circuits. The construction table on page 3 provides eight-step witnesses for every result.

Problem 4. ABABCBADCDA always reduces to A. One order is:
ABABCBADCDA → ABCBADCDA → ABADCDA → ADCDA → ADA → A.
Another begins by removing the B→C→B excursion:
ABABCBADCDA → ABABADCDA → ABADCDA → ADCDA → ADA → A.
Each arrow deletes one adjacent same-road reversal. Different intermediate routes do not imply different final routes.

Problem 5. Write a for left forward (HABH), a^{-1} for left backward (HBAH) and b for right forward (HCDH). Four orders shorten to HCDH: a a^{-1} b, a^{-1} a b, b a a^{-1} and b a^{-1} a, because a ring trip next to its backward trip cancels. The other two, a b a^{-1} = HABHCDHBAH and a^{-1} b a = HBAHCDHABH, contain no out-and-back pair, so they stay as they are and differ. The same three ring trips in a different order can give a different result.

Problem 6. A 12-step disappearing example is HABHBAHCDHDCH: left forward, left backward, right forward, right backward. A 12-step survivor is HABHCDHBAHDCH: left forward, right forward, left backward, right backward. Both use both rings. In the first, nested opposite traversals cancel within each round-trip pair. The second has no immediate reversal anywhere.

Optional hints only when needed

Tree: “What happens when you arrive at a leaf?” Ring: “If you turn around now, did you create a removable pair?” Order: “Where is the first place the two journeys disagree?” Ask for an actual pawn replay to settle a claim that two steps use the same road. Do not cancel distant opposite traversals merely because their directions balance.

Grades 4-5 answer key

Problem 1. Printed results: ABDE, ABDE, CBDF. The seven-step construction CBABDEDF is one valid new C-to-F input with reduced path CBDF. The listed C-to-F input has five steps; the new route must meet the requested seven-step constraint.

Problem 2. No nonempty A-home trip on the tree survives. Root the tree at A. In any nonempty closed journey, take a visited vertex of greatest distance from A. Choose an occurrence reached from its parent. Since no deeper vertex is visited and a tree gives it only one neighbor closer to A, the next step returns along that same edge. There is a cancellable pair. Reapply after each deletion until only A remains.

Problem 3. No. Every C-to-F journey reduces to CBDF. A reduced tree path cannot revisit a vertex, because the subjourney between repeated visits would be a nonempty closed reduced tree walk. It is thus simple, and a tree has only one simple path between the endpoints.

Problem 4. The three printed routes reduce to A, ADC, and ABCDA, respectively. The page 6 proof shows no choice of first reversal can change the final reduced edge path on any graph. Illustrative reductions for the last two:
ABADCBC → ADCBC → ADC.
ABCBCDABA → ABCDABA → ABCDA, or ABCBCDABA → ABCBCDA → ABCDA.
The third route has three possible first removals and four removal orders, all ending at ABCDA. For the first, see the two distinct deletion orders on page 4. These computations illustrate the theorem; they are not its proof.

Problem 5. The requested journey is HABHCDHBAHDCH, or ab a^{-1} b^{-1}. No consecutive pair is an edge followed immediately by that same edge backward, including at H between circuits. Nothing can be removed, so the trip cannot disappear. H is the based endpoint and interior junction; it is not permission to split off or reorder a circuit.

Problem 6. Of the 24 orders of a, b, a^{-1}, b^{-1}, 16 disappear and 8 survive. If two trips on the same ring are next to each other, they are a ring trip and its backward trip, which cancel; the other ring's two trips are then next to each other and cancel too. If the rings alternate, left, right, left, right or right, left, right, left, no out-and-back pair occurs anywhere, including at H: these 8 orders survive (two patterns, each with two ways to place each ring's two trips), such as ab a^{-1} b^{-1}. Every order goes around each ring once each way, so equal net counts do not decide whether a trip disappears; the order does.

Interpretation and limits

This is a graph homotopy problem, not an embedded-rope puzzle. Self-overlap is allowed; moving off a road is not. The two-ring commutator survives here because the graph has no filled face. In the later torus activity, face slides supply additional relations and horizontal/vertical circuits can commute. Do not import those moves into this week.

Why reduction is unique and what is still untested

An invariant stack proof

Read oriented edges from left to right. Keep a stack. If the next edge is inverse to the top, remove the top; otherwise append the edge. The stack remains reduced. To show that this final stack is invariant under deletion anywhere in the input, consider a prefix stack S followed by e and e^{-1}.

If S does not end in e^{-1}, reading e appends it and reading e^{-1} removes it, returning S. If S ends in e^{-1}, write S=T e^{-1}. Reading e removes that last edge. Since S was reduced, T cannot end in e; therefore reading e^{-1} appends it and restores S. Thus inserting or deleting e e^{-1} does not change the stack after that pair, or after the remaining suffix.

Every deletion process terminates because length falls by two. A reduced terminal route reads to itself under the stack rule. The invariant stack is therefore that terminal route for every deletion order. This proves uniqueness for arbitrary finite edge paths and arbitrary graphs when each actual edge and orientation is recorded. It is not an extrapolation from a bounded computer check.

Trees and rings

A tree has a unique simple route between any two vertices: two distinct simple routes would enclose a cycle. The farthest-vertex argument on page 5 rules out a nonempty reduced closed walk. Consequently every reduced tree route is simple and hence is the unique endpoint route. On a ring every vertex has degree two, so a reduced nonempty walk has a fixed direction throughout; its endpoint and length bound give the finite classifications in the keys.

Primary source

Allen Hatcher, *Algebraic Topology*, Chapter 1: §1.1 defines endpoint-fixed path homotopy; §1.2 treats reduced words/free products; §1A, “Graphs and Free Groups,” develops the graph setting. The stack and finite ring arguments here are self-contained versions for the printed road maps. Chapter 1 PDF.

Before-use and observation record

Date and adult: _____
Actual mat and directed-tile/adult-recording workflow tested: _____
Child can replay, delete an adjacent pair, and preserve order: _____
Observed sticking point and adjustment: _____

Status: all packets and this guide are unpiloted. Exact-material tabletop tests and K-1 route-recording readiness have not been performed here. Digital route enumeration verifies listed examples, not physical usability, child readiness, or the arbitrary-length theorem.

Key alignment: K-1 has six pages, Problems 1-7; Grades 2-3 and 4-5 have four pages, Problems 1-6. All three print F39-K-v3, F39-23-v3 and F39-45-v3, revised October 23, 2026. The short ring investigation is K-1 Problem 2; ring continuations are Problems 6-7.

Starting routes and return visits

Planning update of 4 October 2026. The timing menus are examples; a theme can continue over several meetings. Keep the full Grades 4-5 investigations and choose by readiness.

Use and return choices

Keep one adult at each table; the youngest adult can record road names while children choose and replay. First routes: K-1 Problems 1-2, three/four-step tree journeys, then four-step ring home trips; Grades 2-3 Problems 1-4; Grades 4-5 Problems 1-4. Use one ordered card for each actual road traversal; remove only out-and-back pairs and replay the rest. K-1 Problems 3-7 remain for a later visit or when ready. On return, Grades 2-3 and 4-5 take the two-ring order problems (P5-6); for Grades 4-5 these find zero-net-count trips that cannot disappear. The actual tile/mat/pawn procedure remains physically untested.